

SQL JOINS - Visual Guide

How Tables Merge and Rows Match

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Oracle HR Schema - A1 English Level

What is a JOIN?

A JOIN combines rows from two or more tables based on a related column between them. In a database, data is stored in many tables. Each table has information about one thing. But in real life, we often need information from more than one table at the same time. For example, to see an employee name with their department name, we need data from the EMPLOYEES table and the DEPARTMENTS table. A JOIN helps us put these two tables together.

Think of it like this: you have a list of employee names in one paper and a list of department names in another paper. Both papers have a department ID number. A JOIN matches the department ID from the first paper with the department ID from the second paper and puts the information together on one page.

In the HR schema, the EMPLOYEES table has department_id. The DEPARTMENTS table also has department_id. This common column is the "bridge" that connects the two tables. We call this bridge a **foreign key**.

How HR Tables Connect

In the HR schema, tables are connected through shared columns. The diagram below shows the main connections. The department_id column in EMPLOYEES connects to the department_id column in DEPARTMENTS. This is the most common JOIN you will use. Other connections exist too, like manager_id in EMPLOYEES connecting to employee_id in the same table (this is called a Self JOIN).

Table A	Column	Table B	Column
EMPLOYEES	department_id	DEPARTMENTS	department_id
EMPLOYEES	job_id	JOBS	job_id
EMPLOYEES	manager_id	EMPLOYEES	employee_id
DEPARTMENTS	location_id	LOCATIONS	location_id
LOCATIONS	country_id	COUNTRIES	country_id
COUNTRIES	region_id	REGIONS	region_id

This table shows which columns connect which tables. When you write a JOIN, you use these columns in the ON clause. For example: `e.department_id = d.department_id` means "match rows where the department_id in EMPLOYEES equals the department_id in DEPARTMENTS".

Basic JOIN Syntax

Here is the general syntax for a JOIN. The SELECT part says which columns you want to see. The FROM part says the first table. The JOIN part says the second table. The ON part says how the two tables are connected (which columns to match).

```
SELECT table1.column1, table2.column2
FROM table1
JOIN table2
  ON table1.common_column = table2.common_column;
```

The keyword JOIN is the same as INNER JOIN. Oracle also supports table aliases (short names for tables). For example, instead of writing EMPLOYEES every time, you can write 'e'. So the syntax becomes:
e.department_id = d.department_id where 'e' is EMPLOYEES and 'd' is DEPARTMENTS.

Section 1: INNER JOIN

Idea: Only the matching rows from both tables.

INNER JOIN returns only the rows that have a match in BOTH tables. If a row in the left table has no match in the right table, it is NOT shown. If a row in the right table has no match in the left table, it is NOT shown either. Think of it like the middle part of a Venn diagram - only the overlap.

Step 1: Look at the two tables

emp_id	first_name	dept_id
101	Neena	90
102	Lex	90
103	Alexander	60
104	Bruce	60
107	Diana	NULL

dept_id	dept_name
60	IT
90	Executive
99	Unknown

Left Table (green rows = have a match)

Step 2: What the JOIN returns

first_name	dept_id	dept_name
Neena	90	Executive
Lex	90	Executive
Alexander	60	IT
Bruce	60	IT

Result Table (green = matched rows, orange = unmatched rows)

SQL Query on HR Schema:

```
SELECT e.first_name, e.last_name, d.department_name
FROM employees e
JOIN departments d
  ON e.department_id = d.department_id
```

```
ORDER BY e.last_name;
```

What happens in the HR schema: This query shows the employee name and their department name. The JOIN matches department_id in EMPLOYEES with department_id in DEPARTMENTS. Only employees who have a valid department_id are shown. If an employee has NULL department_id, they will not appear in the result.

Key point: INNER JOIN only returns rows that exist in BOTH tables. Department 99 (Unknown) does not appear because no employee works there. Diana (dept_id = NULL) does not appear because NULL cannot match anything.

Section 2: LEFT (OUTER) JOIN

Idea: All rows from the LEFT table + matching rows from the right table.

LEFT JOIN returns ALL rows from the left table, even if there is no match in the right table. If a row in the left table has no match in the right table, the result shows NULL values for the right table columns. This is useful when you want to see ALL employees, including those who do not have a department yet.

Step 1: Look at the two tables

emp_id	first_name	dept_id
101	Neena	90
103	Alexander	60
107	Diana	NULL

dept_id	dept_name
60	IT
90	Executive

Left Table (green rows = have a match)

Step 2: What the JOIN returns

first_name	dept_id	dept_name
Neena	90	Executive
Alexander	60	IT
Diana	NULL	NULL

Result Table (green = matched rows, orange = unmatched rows)

SQL Query on HR Schema:

```
SELECT e.first_name, e.last_name, d.department_name
FROM employees e
LEFT JOIN departments d
  ON e.department_id = d.department_id
ORDER BY e.last_name;
```

What happens in the HR schema: This query shows ALL employees, even those who do not belong to any department. Diana has NULL department_id, so the department_name shows as NULL. But Diana still appears in the result because LEFT JOIN keeps all rows from the left table (EMPLOYEES).

Key point: LEFT JOIN keeps ALL rows from the first table. Unmatched rows get NULL for the second table columns. This is very useful for finding missing data - for example, find all employees who do not have a department: add WHERE d.department_id IS NULL at the end.

Section 3: RIGHT (OUTER) JOIN

Idea: All rows from the RIGHT table + matching rows from the left table.

RIGHT JOIN returns ALL rows from the right table, even if there is no match in the left table. If a row in the right table has no match in the left table, the result shows NULL values for the left table columns. This is useful when you want to see ALL departments, including those that have no employees yet.

Step 1: Look at the two tables

emp_id	first_name	dept_id
101	Neena	90
103	Alexander	60

dept_id	dept_name
60	IT
90	Executive
99	Unknown

Left Table (green rows = have a match)

Step 2: What the JOIN returns

first_name	dept_id	dept_name
Neena	90	Executive
Alexander	60	IT
NULL	99	Unknown

Result Table (green = matched rows, orange = unmatched rows)

SQL Query on HR Schema:

```
SELECT e.first_name, e.last_name, d.department_name
FROM employees e
RIGHT JOIN departments d
  ON e.department_id = d.department_id
ORDER BY d.department_name;
```

What happens in the HR schema: This query shows ALL departments, even those that have no employees. Department 99 (Unknown) appears in the result with NULL for the employee name because no employee works in that department. This helps managers see which departments are empty.

Key point: RIGHT JOIN keeps ALL rows from the second table. Unmatched rows get NULL for the first table columns.

Tip: You can swap the table order and use LEFT JOIN to get the same result. Many developers prefer LEFT JOIN because it is easier to read.

Section 4: FULL OUTER JOIN

Idea: All rows from BOTH tables. Matching rows are joined, unmatched rows show NULL.

FULL OUTER JOIN returns ALL rows from both tables. If a row in the left table has no match in the right table, the right columns show NULL. If a row in the right table has no match in the left table, the left columns show NULL. This JOIN shows everything - nothing is left out.

Step 1: Look at the two tables

emp_id	first_name	dept_id	dept_id	dept_name
101	Neena	90	60	IT
103	Alexander	60	90	Executive
107	Diana	NULL	99	Unknown

Left Table (green rows = have a match)

Step 2: What the JOIN returns

first_name	dept_id	dept_name
Neena	90	Executive
Alexander	60	IT
NULL	99	Unknown
Diana	NULL	NULL

Result Table (green = matched rows, orange = unmatched rows)

SQL Query on HR Schema:

```
SELECT e.first_name, e.last_name, d.department_name
FROM employees e
FULL OUTER JOIN departments d
  ON e.department_id = d.department_id
ORDER BY d.department_name;
```

What happens in the HR schema: This query shows ALL employees and ALL departments. Employees without a department show NULL for department_name. Departments without employees show NULL for employee name. This is the most complete view of the data.

Key point: FULL OUTER JOIN shows everything from both tables. No data is left out. This is useful when you want a complete picture - for example, to find employees with no department AND departments with no employees in one query.

Section 5: CROSS JOIN (Cartesian Product)

Idea: Every row from the left table pairs with every row from the right table. No matching condition is needed.

CROSS JOIN creates every possible combination of rows from both tables. If the left table has 3 rows and the right table has 4 rows, the result has $3 \times 4 = 12$ rows. There is no ON clause because every row matches with every other row. This can create very large result sets, so use it carefully.

Example:

size	color
Small	Red
Medium	Blue
Large	

Result (3 sizes x 2 colors = 6 rows):

size	color
Small	Red
Small	Blue
Medium	Red
Medium	Blue
Large	Red
Large	Blue

SQL Query on HR Schema:

```
SELECT e.first_name, d.department_name
FROM employees e
CROSS JOIN departments d;
```

What happens: Every employee is paired with every department. If there are 107 employees and 11 departments, the result has $107 \times 11 = 1,177$ rows. This is usually not what you want in a real query, but it can be useful for generating all possible combinations.

Key point: CROSS JOIN has no ON clause. It pairs every row with every row. Be careful - if both tables are large, the result can be millions of rows.

Section 6: Self JOIN

Idea: A table is joined with itself. You use two different aliases for the same table.

A Self JOIN is not a special SQL keyword. It is a regular JOIN where the same table appears on both sides. You give the table two different aliases (short names) so Oracle can tell them apart. The most common use of a Self JOIN in the HR schema is to find an employee and their manager. The `manager_id` column in `EMPLOYEES` points to the `employee_id` of another row in the same table.

Example: Employee and their Manager

EMPLOYEES table:

employee_id	first_name	manager_id
101	Neena	NULL
102	Lex	101

103	Alexander	102
104	Bruce	103

In this table, Neena has no manager (she is the CEO). Lex reports to Neena (manager_id = 101). Alexander reports to Lex (manager_id = 102). Bruce reports to Alexander (manager_id = 103). A Self JOIN connects the manager_id to another row in the same table.

Result after Self JOIN:

employee	employee_id	manager	manager_id
Neena	101	(No Manager)	NULL
Lex	102	Neena	101
Alexander	103	Lex	102
Bruce	104	Alexander	103

SQL Query on HR Schema:

```
SELECT e.first_name || ' works for ' || m.first_name
AS "Reporting Line" FROM employees e LEFT JOIN employees m
ON e.manager_id = m.employee_id ORDER BY e.employee_id;
```

What happens: The EMPLOYEES table is used twice with different aliases. "e" is the employee and "m" is the manager. The ON clause connects e.manager_id to m.employee_id. We use LEFT JOIN so that Neena (who has no manager) still appears in the result. The result shows reporting lines like "Lex works for Neena" and "Alexander works for Lex".

Key point: Self JOIN uses the same table twice with different aliases. The most common use case is finding employee-manager relationships. Always use LEFT JOIN for Self JOINs so the top-level person (with no manager) is not excluded.

Section 7: JOIN with WHERE Condition

You can combine JOIN and WHERE in the same query. The JOIN connects the tables, and the WHERE clause filters the results. The WHERE condition is applied AFTER the JOIN, so you can filter on columns from either table.

```
SELECT e.first_name, e.last_name, e.salary, d.department_name FROM employees e JOIN
departments d ON e.department_id = d.department_id WHERE d.department_name = 'IT' AND
e.salary > 5000 ORDER BY e.salary DESC;
```

What happens: First, the JOIN connects EMPLOYEES with DEPARTMENTS using department_id. Then, the WHERE clause filters the results to show only employees in the IT department who earn more than 5000. The ORDER BY sorts by salary from highest to lowest. You can see how the JOIN and WHERE work together: the JOIN brings the data together, and the WHERE selects only the rows you need.

Section 8: Joining More Than Two Tables

You can join more than two tables in one query. To do this, you add more JOIN clauses. Each JOIN connects one table to the next. The order of JOINS matters because each new JOIN can only reference tables that are already in the query.

```
SELECT e.first_name, e.last_name, d.department_name, l.city, l.state_province,  
c.country_name FROM employees e JOIN departments d ON e.department_id = d.department_id  
JOIN locations l ON d.location_id = l.location_id JOIN countries c ON l.country_id =  
c.country_id WHERE e.department_id = 80 ORDER BY e.last_name;
```

What happens: This query joins FOUR tables together. First, EMPLOYEES joins with DEPARTMENTS on department_id. Then, DEPARTMENTS joins with LOCATIONS on location_id. Finally, LOCATIONS joins with COUNTRIES on country_id. The WHERE clause filters to show only Sales department employees (dept 80). The result shows the employee name, department name, city, state, and country - all in one result.

Think of it like a chain: EMPLOYEES -> DEPARTMENTS -> LOCATIONS -> COUNTRIES. Each link in the chain is one JOIN with one ON condition. The chain of connections goes from the employee to their department, then to the office location, and finally to the country.

Section 9: JOIN Types - Quick Comparison

This table summarizes all JOIN types. It shows which rows each JOIN keeps from the left and right tables. Use this table to quickly remember the difference between each type.

JOIN Type	Left Table	Right Table	Unmatched Rows
INNER JOIN	Only matched	Only matched	Excluded from both
LEFT JOIN	ALL rows	Only matched	Right unmatched excluded; left unmatched shown with NULL
RIGHT JOIN	Only matched	ALL rows	Left unmatched excluded; right unmatched shown with NULL
FULL OUTER JOIN	ALL rows	ALL rows	All unmatched shown with NULL
CROSS JOIN	ALL rows	ALL rows	Every row paired with every row
Self JOIN	Same table (alias 1)	Same table (alias 2)	Depends on LEFT or INNER

Section 10: Practice Queries

Try these queries on your own. Each query uses a different JOIN type. Write them in Oracle SQL Developer and check the results.

1. INNER JOIN - Employee with Department

Show first_name, last_name, and department_name for all employees who have a department.

```
SELECT e.first_name, e.last_name, d.department_name  
FROM employees e  
JOIN departments d
```



```
    ON e.department_id = d.department_id  
ORDER BY d.department_name;
```

2. LEFT JOIN - Find Employees with No Department

Show employees who do NOT have a department (department_name will be NULL).

```
SELECT e.first_name, e.last_name, d.department_name  
FROM employees e  
LEFT JOIN departments d  
    ON e.department_id = d.department_id  
WHERE d.department_name IS NULL;
```

3. RIGHT JOIN - Find Empty Departments

Show departments that have no employees.

```
SELECT d.department_name, e.first_name  
FROM employees e  
RIGHT JOIN departments d  
    ON e.department_id = d.department_id  
WHERE e.employee_id IS NULL;
```

4. Self JOIN - Employee and Manager Names

Show each employee name with their manager name.

```
SELECT e.first_name || ' reports to ' || m.first_name  
    AS "Reporting Line"  
FROM employees e  
LEFT JOIN employees m  
    ON e.manager_id = m.employee_id  
ORDER BY e.employee_id;
```

5. Multi-Table JOIN - Employee with Location

Show employee name, department name, city, and country.

```
SELECT e.first_name, e.last_name,  
       d.department_name, l.city,  
       c.country_name  
FROM employees e  
JOIN departments d ON e.department_id = d.department_id  
JOIN locations l ON d.location_id = l.location_id  
JOIN countries c ON l.country_id = c.country_id  
ORDER BY e.last_name;
```

6. JOIN with WHERE - IT Employees with High Salary

Show IT department employees who earn more than 5000.

```
SELECT e.first_name, e.last_name, e.salary,  
       d.department_name  
FROM employees e  
JOIN departments d  
    ON e.department_id = d.department_id  
WHERE d.department_name = 'IT'  
    AND e.salary > 5000
```

```
ORDER BY e.salary DESC;
```

Important Notes

Remember these rules when you write JOIN queries. They will help you avoid common mistakes.

Rule	Explanation
Use table aliases	Always use short names for tables (like "e" for EMPLOYEES, "d" for DEPARTMENTS). This makes your query shorter and easier to read.
Specify columns with alias	When two tables have a column with the same name, always write the alias before the column name. For example: e.department_id, not just department_id.
NULL never equals NULL	A JOIN condition like e.department_id = d.department_id will never match if both values are NULL. That is why employees with NULL department_id do not appear in INNER JOIN results.
Use LEFT JOIN to keep all data	When you want ALL rows from one table, use LEFT JOIN. This ensures no data is lost. You can add WHERE ... IS NULL to find the unmatched rows.
Be careful with CROSS JOIN	CROSS JOIN multiplies the row count. If table A has 100 rows and table B has 50 rows, the result has 5,000 rows. Always check the row count before running a CROSS JOIN.
Column aliases use double quotes	In Oracle, column aliases with spaces need double quotes: AS "Full Name". Single quotes are only for string values like 'IT'.